

CASTELLS' KEY IDEAS

1 NETWORK SOCIETY

This information age has never been a technological matter. It has always been a matter of social transformation, a process of social change in which technology is an element that is inseparable from social, economic, cultural and political trends.

(Castells with Catterall 2001: 3)

In order to fully grasp the forms and functions of the network society, we need to attend to the theoretical building blocks and witness the moves that Castells makes in assembling his argument. These are extensively elaborated in *The Rise of the Network Society* (1996/2000) and succinctly summarized in a series of articles (see especially Castells 2000b, 2001a, 2002 [1996]).

INFORMATIONALISM

First is the concept of informationalism, which Castells uses to delineate a new economic system and era, preferring the term over other epochdefiners such as 'post-industrialism'. He also prefers to think of the transition to an informational economy in terms of a socio-technical 'paradigm shift' rather than a revolution (Castells 2001a). As global competitiveness (of workers, corporations, states) increasingly depends on access to and ability to manipulate information, so we have a new economy: 'in the new, informational mode of development the source of productivity lies in the technology of knowledge generation, information processing, and symbol communication. ... [T]he action of knowledge upon knowledge itself [is] the main source of productivity' (Castells 1996a: 17). As Webster and Robins (1998) put it, this entails a shift in production from hardware to software and to data. And given the fatal collapse of statism (communism), we have in fact entered the age of global informational *capitalism*, which Castells only half-jokingly names 'a new brand of capitalism' (Castells 2000a: 52).

CAPITALISM

An economic system in which the means of production are owned and controlled privately, labour is exchanged for wages, and the production, distribution and

pricing of goods and services are determined by supply and demand in a market context, rather than by the state. Those in control of the means of production generally run them for profit, which is appropriated by the owners of capital – the capitalists. Capitalism contrasts with socialism or communism, where the means of production and the commodities produced are owned and used by the state or collectively. An economy with a large amount of state intervention combined with capitalist characteristics is often referred to as a mixed economy and, if state intervention is dominant, as statism.

At the heart of informationalism is what Castells refers to as the informational mode of development, a term he uses to distinguish what's changed in this new economy: we are still in a capitalist mode of production, in terms of the relations between capital, property ownership and labour (even if, as we shall see, he argues that the social structure, i.e. class, has fundamentally changed). Following his old mentor, Alain Touraine, he specifies a mode of development as 'the technological arrangements through which humans act upon matter (nature), upon themselves, and upon other humans' in order to generate wealth (Castells 2000b: 9). Modes of development are defined, he adds, 'by their central technological paradigm and by their principle of performance' (ibid.), hence the shift from an industrial to an informational mode of development: wealth is now created, under a capitalist mode of production, through the action of knowledge upon knowledge. Here is how Castells lays out the bigger picture:

The emergence of a new technological paradigm organized around new, more powerful, and more flexible information technologies makes it possible for information itself to become the product of the production process. To be more precise: the products of new information technology industries are informationproducing devices or information processing itself. New information technologies, by transforming the processes of information processing, act upon all domains of human activity, and make it possible to establish endless connections between different domains, as well as between different elements and agents of such activities. A networked, deeply interdependent economy emerges that becomes increasingly able to apply its process in technology, knowledge, and management to technology, knowledge, and management themselves.

(Castells 1996a: 67)

So the economy is informational, its elements are deeply interdependent, and it is also global: this does not equate to a 'world economy', but means that we now have an economic system 'whose core,

strategically dominant activities have the potential of working as a unit in real time on a planetary scale’ (Castells 2002 [1996]: 126) – exemplars include financial markets and multinational corporations.

Financial globalization, which Castells concentrates his sights on for a long time and which he sees as ‘the major structure of domination in our world today’ (Castells with Catterall 2001: 29), has become so disembedded and disconnected from ‘the ground’ that it functions almost autonomously; Castells refers to it, in one of his rare *tech-noir* moments, as the Automaton:

We have created an Automaton, at the core of our economies, decisively conditioning our lives. Humankind’s nightmare of seeing our machines taking control of our world seems on the edge of becoming reality – not in the form of robots that eliminate jobs or government computers that police our lives, but as an electronically based system of financial transactions.

(Castells 2000a: 56)

This Automaton, which he also refers to as a faceless ‘collective capitalist’, is not just a robotized market, however; its logic is more complex and chaotic, even random – the financial system is marked by ‘information turbulences’, unforeseen booms and busts, such as those around so-called [dot.com](#) businesses. The chaotic dynamics of the global economy, moreover, can make their effects felt anywhere and everywhere, purposively or incidentally, even though large parts of the world are formally excluded from its operations. Castells refers to these excluded regions as the Fourth World, places skipped over or stripped out by global finanscapes (Appadurai 1990). As Castells makes clear, this new economy has a profoundly uneven cartography (but with what he calls an enduring architecture): ‘the global economy emerging from informational-based production and competition is characterized by its *interdependence*, its *asymmetry*, its *regionalization*, the *increasing diversification within each region*, its *selective inclusiveness*, its *exclusionary segmentation*, and, as a result of all these features, an extraordinarily *variable geometry* that tends to dissolve historical, economic geography’ (Castells 1996a: 106, emphasis in original). These features are manifest at a variety of spatial scales, from nation-states to regions, organizations to workers.

APPADURAI’S SCAPES

According to anthropologist Arjun Appadurai (1990), the new global cultural economy is characterized by disjunctive flows, or ‘scapes’. He identified five:

- ethnoscapescapes – flows of *people*: migrants, refugees, tourists, guest workers—moving individually or collectively, permanently or temporarily, voluntarily or involuntarily
- technoscapescapes – flows of *machinery*: mechanical and informational, low tech and high tech, including technical skills
- finanscapescapes – flows of *money* or capital—both legitimate and illicit, global stock markets and money laundering
- mediascapescapes – flows of *images*: capabilities to produce and disseminate media ‘texts’, or what Appadurai calls ‘image-centred, narrative-based accounts of strips of reality’ that take on different meanings depending on where they are consumed
- deoscapescapes – flows of *ideas* or ideologies: especially the globalization of Western metanarratives such as democracy, freedom, human rights, etc.

NETWORKING

So, this new economy requires new ways of working, too, embodied for Castells in the network enterprise and the networker (someone who works by networking, and also who works in the net). The network enterprise is a flexible arrangement suitable to the needs of informational capitalism; it is defined by the networking logic with its antecedents in the crisis of capitalism, which was partly resolved by restructuring how business organizations work (as well as by extensive deregulation). So-called post-Fordist flexible specialization replaced Fordist mass production; new work cultures emphasized worker autonomy and multitasking (‘Toyotism’); the small firm became a role model for entrepreneurialism and ‘enterprise culture’; and ultimately the network enterprise emerged as the ideal organizational form – a flexible association of components, or nodes, that is never still, but changes from project to project: ‘networks are the fundamental stuff of which new organizations are and will be made’ (Castells 1996a: 168). States, too, begin to look like networks, either by devolving to regions or by linking up strategically in international pacts, with other states and with organizations. Networks are horizontal, nonhierarchical, fluid and mobile, and their unit of work is the project.

The flexible structure of the network is governed by the simplest binary logic: on or off, inclusion or exclusion: networks are ‘value-free. They can equally kiss or kill: nothing personal. It all depends on the goals of a given network and on its most elegant, economical, and self-reproductive form to perform these goals’ (Castells 2001a: 167). So, what is the impact of the network form of business organization on patterns and experiences of employment? Basically the same binary logic: in or out, switched on or

switched off. There is a new (Castells says newest) division of labour in the global informational economy, with four different types or classes of labour: ‘the producers of *high value*, based on informational labor; the producers of *high volume*, based on lower-cost labor; the producers of *raw materials*, based on natural endowments; and the *redundant producers*, reduced to devalued labor’ (Castells 1996a: 147, my emphasis). At the same time, and potentially adding confusion, Castells says that there are three positions available in the new global occupational structure, which he labels networkers, jobless and flextimers.

POSTINDUSTRIALISM

Describes shifts in the economy and society as they move from a manufacturing industrial base towards service, knowledge and information sectors. Associated with sociologist Daniel Bell’s (1974) *The Coming of the Post-industrial Society*, the concept has blurred into other, related notions such as post-Fordism and flexible specialization, all of which describe the political, social, economic and spatial transformations brought about by the decline in the economic importance of manufacturing and its effects on relations of production and consumption.

The language used to describe the new world of work is familiar from ‘management speak’: downsizing, delayering, multitasking, outsourcing, subcontracting, portfolio careers – all euphemisms for the dramatic changes in the organization of work in this new economy that have brought new insecurities and anxieties for many. The lucky ones get to be ‘networkers’ at the top of the pile. Whole labour forces meanwhile can be ‘switched off’, bypassed, routed round: as Castells puts it, capital is global but this labour is local, unlike the footloose nomads of the transnational service class, the networkers. De-unionization, casualization, marketization, individualization, feminization – these are the fates facing flextimers hoping not to be switched off.

Flextimers are often casually employed on a contract-by-contract basis, negotiated via agencies: thus they lack security of employment and attendant benefits; they are required to ‘flex’ their time, to work in patterns suited to the needs of their employers, who may pay them piece rates and who ‘incentivize’ productivity through surveillance; they are often underemployed, working part-time. Often this kind of work recruits female labour; as Castells (2001a: 171) observes, ‘the organization man [is] now replaced by the flexible woman’. This group, who Castells also labels ‘generic labour’, is seen as disposable, replaceable either with more compliant human labour from somewhere else, or by even more compliant

nonhuman labour in the form of automation. These workers are networked rather than networkers, and can be seen and treated as little more than ‘human robots’ (Castells 1996a: 244). So there is a ‘stepped up division of labour’ with, on the one hand, ‘self-programmable labour’ – networkers well equipped to adapt to the demands of the flexible information workplace; and, on the other, ‘generic labour’, exchangeable and disposable. Further beyond that there are ‘legions of discarded, devalued people [who] form the growing planet of the irrelevant’ and whose only entry into global info-capitalism is through the ‘perverse connections’ of the global criminal economy (Castells 2000b: 12).

NETWORK CULTURE

Castells spends a fair amount of time analysing the work and life patterns of networkers, joining a lineage of scholars seeking to understand these new transnational, cosmopolitan technocratic - financial - managerial nomads (Webster 2002). The characteristics of this top strata of ‘informational labour’, seen by Castells and other commentators as powering the new economy and comprising the ‘collective capitalist’, include their ‘footlooseness’ – they feel no sense of rootedness to particular places, no strong sense of local or national identity, nor any ‘organizational identity’ based on belonging to a particular firm. They have highly transferable skills, enabling them to adapt quickly and smoothly to ever-changing networks and projects – they can, in management parlance, multitask (remember that Castells calls them ‘self-programmable labour’; they don’t need training or reskilling, because they constantly upskill themselves). They are not white collar, but, as Andrew Ross (2003) puts it, ‘no collar’ – casual in attire and attitude, but deadly serious about their work. They are as restless as the flows of the financial markets, yet they are still concentrated in major metropolitan centres, where they carve out cosmopolitan lifestyles through their consumption practices. They live in key nodes of what Castells calls the ‘space of flows’ (see below).

There exists, Castells argues, a worldwide network of exclusive enclaves, which comprise the habitat of this elite: boutique hotels and loft apartments, VIP lounges at airports, exclusive restaurants, personal trainers, high-culture events. Castells is even able to cartoon the components of this group’s lifestyle:

the regular use of spa installations (even when traveling), and the practice of jogging; the mandatory diet of grilled salmon and green salad, with *udon* and *sashimi* providing a Japanese functional equivalent; the ‘pale chamois’ wall color intended to create the cozy atmosphere of the inner space; the ubiquitous laptop computer; the combination of business suits and sportswear; the unisex dressing style; and so on.

(Castells 1996a: 417)

As we shall see, this group has made its mark on the cities it inhabits or passes through, too, in terms of new building types and architectural and interior styles that are serially reproduced to make these nomads feel (temporarily) at home, wherever they land.

Now, there is a further important dimension to add to the mix here: what is the defining work ethic of the new information economy? What, Castells asks, glues together the networks of networkers? Unsurprisingly, he names it, in an echo of Max Weber, the spirit of informationalism. Here Castells begins to formulate ideas about the culture of the network society, first off in terms of its organizational codes: it is, he says, ‘a culture of the ephemeral, a culture of each strategic decision, a patchwork of experiences and interests, rather than a charter of rights and obligations. It is a *multifaceted, virtual culture*’ (Castells 1996a: 199, emphasis in original).

BEYOND THE NETWORK

On the periphery of and outside these networks, as we have seen, there are other workers, unable to share in this virtual culture or to surf through the space of flows. The duality between the two positions is stark, Castells argues, and becoming ever starker: ‘extraordinary creativity and extraordinary social inequality and social exclusion are going hand in hand’ (Castells with Catterall 2001: 4). What (limited) options are available to those switched-off populations unable to reap the benefits of the network society? One option, as we have already seen, is to participate in the ‘perverse connections’ of the global criminal economy, to patch in to other networks that work with the same logic but with very different means and ends. Another is withdrawal, retreat into what Castells calls ‘cultural communes’ built around assorted fundamentalisms: ‘there is life beyond the network society: in the fundamentalist cultural communes that reject dominant values and build autonomously the sources of their own meaning; sometimes around self-constructed alternative utopias; more often, around the transcendent truths of God, Nation, Family, Ethnicity, and Territoriality’ (Castells 2001a: 170).

In addition, there are possibilities to appropriate the network society, to make its logic work for other ends, in a process Castells (1999) calls ‘grassrooting the space of flows’ in echo of his earlier work on social movements. These movements may intersect in complex ways with the space of flows, with the global criminal economy, and with cultural communes, as his studies of the American patriot movement, Aum Shinrikyo, the Zapatistas, al-Qaeda, the anti-globalization and green movements, and feminist and lesbian and gay politics all show in different ways (Castells 1997/2004). But Castells is overall quite sanguine about this new social morphology: ‘a society made up of the juxtaposition of flows and tribes ceases to be a society’, he writes (Castells 1996b: 31), noting that social movements can be regressive as well as progressive – qualifying his earlier statement that social movements were the only remaining

engines of social change in the network society (see also Castells and Ince 2003). But this position is seen as fatalistic by some critics, in that it suggests that global informational capitalism has ‘won’ and has ‘switched off’ any alternatives (Webster and Robins 1998). He has continued to work over this part of his analysis, but his final view on the grassroots seems ambivalent.

NETWORK SELF

In *The Internet Galaxy*, Castells ponders some more the ‘society’ in network society, siding with those researchers and critics who see the ‘social reality’ of the Internet, in terms of users’ participation, as ‘an extension of life as it is, in all its dimensions, and with all its modalities’ (Castells 2001b: 118). Reviewing the available evidence, and not believing the hype, he ends up ambivalent here, too, about the effects of Internet use on social relations and sociability, even while he sees sociability being transformed off-line. However, his reading of online social interaction leads him to formulate the idea of ‘networked individualism’, noting that virtual communities are ‘me-centered networks’ or ‘personalized communities’ wherein sociability is ‘privatized’ (ibid.:128; see also Wellman and Haythornwaite 2002). As noted above, the expanding and proliferating platforms for individuals to produce and consume (or ‘prosume’) mediated reflections of self-identity, from blogging to podcasting to on-line gambling (see below), provide endless opportunities for the exercise of networked individualism. Moreover, this individualism, Castells argues, is the result of the network society as sketched above: it is the outcome of the individualization of work in the network enterprise, of the decline of patriarchalism, of withdrawal from politics and civil society, and of new patterns of urbanization and the dislocation of the spaces of everyday life in cities cut through by flows (see below). Individuals caught in the space of flows become networks themselves, and networked individualism becomes the new social pattern (Castells is keen to stress this is not merely a collection of isolated individuals). This is in some ways a transformation of his earlier discussion of the dialectic, even conflict, between the net and the self (e.g. Castells 1996a / 2000); here the net *is* the self, and people have assembled ‘portfolios of sociability’ to match their portfolio careers, ‘project identities’ to match their project-based work lives: ‘the construction / reconstruction of the self is tantamount to managing the changing set of flows and codes that people are confronted with in their daily existence’ (Castells 1996b: 34). Or, as he puts it in an autobiographical aside, ‘I am afraid I look more like a flow’ (Castells 1996a / 2000: 243).

SUMMARY

The network society has a number of distinguishing features, though critics are unsure just how ‘new’ this all is (Castells says the question of newness is boring). We are now living under the condition of global informational capitalism, brought about by three independent processes: the information technology ‘revolution’, the crisis and restructuring of capitalism and statism, and the rise of new social movements. Out of this whirl of change come the network society and the network logic. The global economy is chaotic and turbulent, and seems to function autonomously, with very variable impacts around the world. There is a new spatial division of labour, based around self-programmable elite labour, replaceable generic labour, and the discarded jobless. Working now means networking, and organizations become network enterprises while individuals develop portfolio careers or risk being switched off. The elites of self-programmable labour live exclusive lifestyles while social exclusion and poverty escalate around them; the poor and excluded react by joining the global criminal networks, or by withdrawing into fundamentalisms. Networking logic extends to the self, either by counterposing the net and the self or through network individualism, though grassroots offers ways to bring together the net and the self, through the networking of project identities.

2 SPACE OF FLOWS

Castells says that his concept of the space of flows is very important and very complex, even counter-intuitive. First elaborated in *The Informational City*, it is one area of *The Rise of the Network Society* where he permits himself a more theoretical approach (and later even admits to previous theoretical mistakes; see Castells and Ince 2003). Based on the observation that ‘dominant functions were increasingly operating on the basis of exchanges between electronic circuits linking up information systems in different locations’, Castells’ central premise is the dichotomy between the space of flows and the space of places as the geographical manifestations of the network society (Castells 2002 [1996]: 131). The space of flows is woven closely together with his idea of ‘timeless time’, which I also discuss in this section.

NETWORKS AND FLOWS

The notion of flows is integral to the idea of the network:

